

RETAILSPHERE (PTY) LTD

Data Platform Programme

Functional Requirements Document

Step 3 Deliverable — What the Platform Must Do

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Total Requirements	36 requirements across 6 marts and 8 cross-cutting
Status	Draft — Pending CDO Sign-off

1. Purpose

This document defines the functional requirements for the RetailSphere Data Platform. It answers one question — what must the platform do? Requirements are derived directly from Step 2 Business Process Analysis stakeholder interviews and are organised into six data mart areas plus cross-cutting platform requirements.

Every requirement in this document traces back to a specific stakeholder need documented in BPA-001. The traceability matrix in Section 11 makes this connection explicit.

2. Scope

This document covers functional requirements only — what the platform must do. Non-functional requirements — how fast, how secure, how available — are documented separately in NFR-001 (Step 4).

Mart	Domain	Process Owner	Requirements
Mart 1	Procurement Analytics	Sipho Dlamini	FR-PRO-001 to FR-PRO-005
Mart 2	Inventory Management	Zanele Mokoena	FR-INV-001 to FR-INV-005
Mart 3	Sales Analytics	Rethabile Sithole	FR-SAL-001 to FR-SAL-003
Mart 4	Customer Analytics	Priya Naidoo	FR-CUS-001 to FR-CUS-006
Mart 5	Finance and GL	Johan van der Merwe	FR-FIN-001 to FR-FIN-005

Mart	Domain	Process Owner	Requirements
Mart 6	Marketing Attribution	Rethabile Sithole	FR-MKT-001 to FR-MKT-004
Cross-cutting	All Domains	T. Mbalo / N. Mahlangu	FR-CCR-001 to FR-CCR-008

3. Requirement Format and Conventions

Each requirement is documented using a standard six-field format:

- Requirement ID — unique identifier in format FR-[MART]-[NUMBER]
- What — the business capability the platform must deliver
- How Measured — the calculation or logic applied
- Frequency — how often the data is updated or calculated
- Source Systems — where the data originates
- Acceptance Criteria — how we confirm the requirement is met

Where applicable, requirements also include Dependencies — ADRs or other requirements that must be resolved first — and Notes for important constraints or findings.

4. Mart 1 — Procurement Analytics

Process Owner: Sipho Dlamini, Head of Procurement. Source: BPA-001 Section 3.1.

Sipho requires five metrics available every Monday morning without manual calculation. Currently his team spends three days every month producing the procurement savings number alone. All five requirements depend on SAP ECC as the primary source. FR-PRO-001 and FR-PRO-003 additionally depend on ADR-003 supplier contract PDF extraction.

FR-PRO-001 — Supplier On-Time Delivery Rate

Requirement ID	FR-PRO-001
What	Calculate on-time delivery rate per supplier across all 340 active suppliers
How Measured	Actual GRN delivery date compared to contracted lead time from supplier contract master. Expressed as percentage of orders delivered on or before agreed date. Average days late shown for orders that missed the deadline.
Frequency	Daily
Source Systems	SAP ECC GRN records (actual delivery) + Supplier contract master extracted via ADR-003
Acceptance Criteria	Report shows on-time delivery rate for each of 340 active suppliers. Sipho can sort by worst performing supplier. Platform result matches manual calculation within 0.1% on first validation run.
Dependencies	ADR-003 — Supplier Contract PDF Extraction must be resolved before contracted lead time data is available

FR-PRO-002 — Invoice Accuracy Rate

Requirement ID	FR-PRO-002
What	Calculate invoice rejection rate per supplier including rejection reason per invoice
How Measured	Failed three-way match count and rejection reason per invoice. Expressed as percentage of total invoices submitted per supplier.
Frequency	Daily
Source Systems	SAP ECC invoice matching records
Acceptance Criteria	Platform shows all 847 current exception queue invoices with rejection reason and supplier. Count matches current Excel exceptions queue exactly on first validation run.

FR-PRO-003 — Price Variance Against Contracted Rates

Requirement ID	FR-PRO-003
What	Calculate variance between contracted price and actual amount paid per supplier
How Measured	Actual SAP payment amount compared to contracted rate extracted from supplier contract PDFs. Variance expressed in rands and as a percentage.
Frequency	Weekly
Source Systems	SAP ECC actual payment records + Supplier contracts extracted via ADR-003
Acceptance Criteria	Sipho confirms platform variance matches his manual calculation using SAP payments and contract PDFs as source. Variance visible per supplier and per product category.
Dependencies	ADR-003 — Supplier Contract PDF Extraction must be resolved before contracted rates are available. This requirement cannot be built until ADR-003 is signed.

FR-PRO-004 — Days Payable Outstanding

Requirement ID	FR-PRO-004
What	Calculate days payable outstanding per supplier expressed in business days
How Measured	Invoice due date versus current date. Calculated in business days excluding weekends. Grouped by supplier and by payment terms category — 30, 60, and 90 day terms.
Frequency	Daily
Source Systems	SAP ECC invoice dates + SAP payment run records
Acceptance Criteria	Platform shows DPO per supplier in business days. Result matches SAP creditors report on first validation day.

FR-PRO-005 — Procurement Savings as Percentage of Total Spend

Requirement ID	FR-PRO-005
What	Calculate procurement savings as a percentage of total spend
How Measured	List price minus actual paid divided by total spend. Expressed as a percentage.
Frequency	Monthly
Source Systems	SAP ECC purchase order values + Market benchmark data

Acceptance Criteria	Currently takes Sipho's team three days of manual Excel calculation. Platform delivers same number in seconds. Sipho confirms result matches manual calculation on first validation run.
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5. Mart 2 — Inventory Management

Process Owner: Zanele Mokoena, National Inventory Manager. Source: BPA-001 Section 3.2.

Zanele's requirements split across two fact tables. FR-INV-001 to FR-INV-003 are served by the Inventory Periodic Snapshot. FR-INV-004 and FR-INV-005 are served by the Replenishment Accumulating Snapshot. BPA-FINDING-001 applies — inventory metrics will be directionally correct but quantitatively approximate until three structural operational gaps are resolved.

FR-INV-001 — Stock Position per SKU per Store

Requirement ID	FR-INV-001
What	Current stock quantity per SKU per store across all 87 stores
How Measured	Quantity on Hand from Inventory Periodic Snapshot. Semi-additive measure — can be averaged across dates but not summed.
Frequency	Daily — real-time not achievable given structural data gaps in shrinkage, returns, and inter-store transfers. BPA-FINDING-001 applies.
Source Systems	SAP ECC stock management records + Oracle MICROS POS real-time decrements
Acceptance Criteria	Zanele can compare platform stock position per SKU per store against SAP and POS figures. Discrepancies flagged for investigation. Numbers are directionally correct — formally acknowledged as approximate pending BPA-FINDING-001 resolution.
Note	BPA-FINDING-001 — Inventory domain structural data integrity risk. Business acknowledgement required before go-live.

FR-INV-002 — System vs Physical Stock Variance

Requirement ID	FR-INV-002
What	Variance between system stock and physical count per store expressed in units and percentage
How Measured	SAP system stock quantity minus physical count submitted by each store. Variance expressed in units and as a percentage of total stock.
Frequency	Weekly — physical count data arrives via email from 87 stores every Monday morning
Source Systems	SAP ECC system stock records + Manual physical stock count data submitted as Excel per store every Monday
Acceptance Criteria	Zanele can compare platform variance report against her current Monday Excel and email submissions. Count and percentage must match on first validation run.

FR-INV-003 — Shrinkage Rate by Product Category

Requirement ID	FR-INV-003
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What	Shrinkage rate per product category per store expressed as percentage of total stock value
How Measured	Shrinkage units and value expressed as percentage of total stock value per category per store.
Frequency	Weekly — shrinkage data arrives via email from 87 stores every Monday morning
Source Systems	SAP ECC stock records + Shrinkage reports submitted as Excel per store every Monday
Acceptance Criteria	Zanele can compare platform shrinkage numbers per store and category against current Excel spreadsheets. Must match on first validation run.

FR-INV-004 — Open Replenishment Orders by Pipeline Stage

Requirement ID	FR-INV-004
What	All open replenishment orders visible by store, SKU, and current pipeline stage in real time
How Measured	Replenishment Accumulating Snapshot. Pipeline stages: Reorder Date, DC Pick Date, Dispatch Date, Store Receipt Date. Status derived from date columns — Open, Picked, In Transit, Received.
Frequency	Real-time — updated as each pipeline stage is completed
Source Systems	Replenishment Accumulating Snapshot fact table
Acceptance Criteria	Zanele can track and confirm progress of any open replenishment order with store, DC, and delivery team. Replaces manual Monday morning spreadsheet currently rebuilt in three hours every week.

FR-INV-005 — Average Days in Replenishment Pipeline

Requirement ID	FR-INV-005
What	Average days from reorder to store receipt per store, DC, and product category
How Measured	Days in Pipeline stored measure from Replenishment Accumulating Snapshot. Includes weekend exclusion logic per Kimball Chapter 6. Calculated from Reorder Date to Store Receipt Date.
Frequency	Daily
Source Systems	Replenishment Accumulating Snapshot — Days in Pipeline stored measure
Acceptance Criteria	No existing baseline to compare against — this metric does not currently exist at RetailSphere. Nomvula Mahlangu to review and approve the calculation methodology before platform goes live. Zanele to confirm output is operationally sensible based on domain knowledge.

6. Mart 3 — Sales Analytics

Process Owner: Rethabile Sithole, Chief Commercial Officer. Source: BPA-001 Section 3.3.

All three Sales requirements depend on ADR-002 Unified Customer Identity Resolution. Without a resolved customer identifier, CLV and retention calculations will only reflect single-channel activity and will not represent the true cross-channel customer picture.

FR-SAL-001 — Customer Lifetime Value Across All Channels

Requirement ID	FR-SAL-001
What	Customer Lifetime Value per customer across all channels — not per channel
How Measured	Total spend per unified customer across POS transactions, Shopify orders, and loyalty-linked sales. Single CLV figure per customer regardless of channel.
Frequency	Daily
Source Systems	Oracle MICROS POS + Shopify Plus + Salesforce CRM
Acceptance Criteria	Rethabile can trace any customer and see their total spend across all channels. Platform shows CLV for all 2.1 million loyalty members plus anonymous segment estimates. Numbers reconcile to individual channel totals on first validation.
Dependencies	ADR-002 — Unified Customer Identity Resolution must be resolved before cross-channel CLV can be calculated

FR-SAL-002 — True Retention and Churn Rate

Requirement ID	FR-SAL-002
What	True retention and churn rate across all channels using agreed six-month engagement definition
How Measured	Platform calculates last engagement date per customer across purchase, browse, email open, and store visit signals. Six-month no-engagement threshold agreed with Rethabile. Customer with no signal in six months flagged as lapsed.
Frequency	Daily data refresh. Monthly churn rate calculation and reporting.
Source Systems	Oracle MICROS POS + Shopify Plus + Salesforce CRM
Acceptance Criteria	Rethabile can see retention rate and churn rate monthly. Can drill into lapsed customers by last engagement date and last engagement channel. Single churn definition used consistently across all channels.
Dependencies	ADR-002 — Unified Customer Identity Resolution required to link engagement signals across all channels to the same person

FR-SAL-003 — Customer Segment Framework

Requirement ID	FR-SAL-003
What	Four customer segments calculated and refreshed monthly with full SCD Type 2 history
How Measured	Platform classifies each customer into one of four segments: Anonymous in-store (no identity, 66% of store transactions); Online only (Shopify, no loyalty link); Loyalty only (no online activity); Full cross-channel (linked across all three systems). SCD Type 2 tracks segment attribute history.
Frequency	Monthly refresh
Source Systems	Oracle MICROS POS + Shopify Plus + Salesforce CRM
Acceptance Criteria	Rethabile can see segment size, revenue contribution, and movement between segments month on month. SCD Type 2 shows where each customer started, when they moved, and what triggered the move.
Dependencies	ADR-002 — Unified Customer Identity Resolution required for cross-channel and loyalty segment classification

7. Mart 4 — Customer Analytics

Process Owner: Priya Naidoo, Head of CRM and Loyalty (Data Steward — Customer dimension). Source: BPA-001 Section 3.4.

Priya owns the Customer dimension as Data Steward. FR-CUS-002 is a POPIA legal obligation — not a preference. FR-CUS-001 enforces the four-contact monthly cap agreed with Priya. ADR-007 must be signed before any contact governance logic is built.

FR-CUS-001 — Communications per Member

Requirement ID	FR-CUS-001
What	All communications sent to each member visible in one place. Maximum four contacts per member per month enforced across all channels combined.
How Measured	Contact History fact table — one row per communication event per member per channel. Rolling 30-day contact frequency calculated from this table. Platform suppresses sends that would breach the four-contact monthly cap.
Frequency	Daily
Source Systems	Contact History fact table sourced from Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4
Acceptance Criteria	Priya can see total contact attempts per member, which campaign each communication belonged to, and which campaigns generated highest unsubscribe rates. Platform enforces four-contact monthly cap and confirms suppression when cap is reached.
Dependencies	ADR-007 — Contact Governance and Opt-out Propagation Architecture must be signed before this is built

FR-CUS-002 — Opt-out Propagation

Requirement ID	FR-CUS-002
What	Member opt-out propagated across all four systems within four hours of opt-out event
How Measured	Tiered ingestion pipeline — near real-time, maximum four hours from opt-out event to full cross-system propagation across Salesforce CRM, Marketing Cloud, Meta, and GA4.
Frequency	Near real-time — hourly batch minimum. Four-hour maximum.
Source Systems	Salesforce CRM + Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4
Acceptance Criteria	Priya can confirm that a member who unsubscribed from one channel is excluded from all four systems within four hours. Platform sends confirmation alert to Priya's team when propagation is complete.
Dependencies	ADR-007 — Contact Governance Architecture must be signed. POPIA legal obligation — this is non-negotiable.
Note	POPIA compliance requirement. This is a legal obligation not a preference. Breach constitutes a regulatory risk.

FR-CUS-003 — At-risk Member Flag

Requirement ID	FR-CUS-003
What	Members at risk of lapsing flagged before they leave using three-component calculation
How Measured	Three components calculated per member daily: last purchase date (from Sales fact table); last category purchased (derived from Sales fact table joined to Product dimension); last engagement date (from Contact History fact table). Members with no engagement signal in six months flagged as at-risk.
Frequency	Daily
Source Systems	Contact History fact table + Sales fact table + Product dimension
Acceptance Criteria	For every at-risk member Priya's team can see last purchase date, last category purchased, and last engagement date. Team can intervene before member lapses — not after.
Dependencies	ADR-002 — Unified Customer Identity required to link engagement signals across all channels to the same member

FR-CUS-004 — Campaign Performance Dashboard

Requirement ID	FR-CUS-004
What	Automated campaign performance dashboard combining all three marketing systems in one view
How Measured	Contact History fact table unified with Salesforce CRM and Marketing Cloud data. Replaces manual reports pulled from three separate systems.
Frequency	Weekly
Source Systems	Contact History fact table + Salesforce CRM + Salesforce Marketing Cloud
Acceptance Criteria	Priya can compare automated dashboard output against current manual campaign reports from three separate systems. Dashboard shows performance for all 2.1 million registered members. Numbers reconcile on first validation run.

FR-CUS-005 — Net Member Movement and Predicted Churn

Requirement ID	FR-CUS-005
What	Net member movement monthly plus predicted churn for next 30 days
How Measured	Net movement: new members minus lapsed members per month. Predicted churn: members approaching six-month no-engagement threshold identified from Contact History fact table rolling calculation.
Frequency	Monthly calculation. Daily data refresh underlying the calculation.
Source Systems	Salesforce CRM + Sales fact table + Contact History fact table
Acceptance Criteria	Priya can see net member movement — new, retained, and lapsed — for the current month. Predicted churn list shows members likely to lapse in next 30 days with enough lead time to intervene.

FR-CUS-006 — Customer Segment History

Requirement ID	FR-CUS-006
What	Track customer movement between four segments over time using SCD Type 2

How Measured	Customer dimension segment attribute tracked with previous segment, current segment, date of change, and trigger that caused the move. SCD Type 2 preserves full history.
Frequency	Monthly refresh — aligned to segment framework monthly update
Source Systems	Customer dimension segment attribute + Sales fact table (purchase triggers) + Contact History (engagement triggers)
Acceptance Criteria	Platform shows where a customer started, when they moved, and what triggered the move. Rethabile can understand what converts anonymous customers to loyalty members and systematise it.
Dependencies	ADR-002 — Unified Customer Identity required for accurate cross-channel segment classification

8. Mart 5 — Finance and GL

Process Owner: Johan van der Merwe, Group Financial Controller. Source: BPA-001 Section 3.5.

Johan carries personal liability for the accuracy of financial statements. His instruction is recorded: the audit trail is section one, not section five. The reconciliation framework underpins every Finance requirement — if the platform number does not match SAP to the cent the pipeline stops before Gold layer is updated. Two consecutive months of parallel running with Excel are required before Excel is decommissioned — Johan's sign-off required each month.

FR-FIN-001 — Audit Trail Report

Requirement ID	FR-FIN-001
What	Complete audit trail of every number, transformation, business rule, and change in the platform — exportable to PDF for auditors
How Measured	dbt transformation logs capture every model run, every rule applied, and every change made. Version controlled in Git — no silent changes possible. Exportable to PDF on demand.
Frequency	Available on demand within 24 hours of any auditor request. Monthly scheduled run for proactive audit readiness.
Source Systems	dbt transformation logs + SAP ECC + Bronze, Silver, and Gold layer metadata
Acceptance Criteria	Johan can see every number, transformation, business rule, and change. Can export to PDF within 24 hours of auditor request. Audit trail is section one — accessible immediately without navigating to a secondary report.

FR-FIN-002 — General Ledger Reconciliation

Requirement ID	FR-FIN-002
What	Platform GL number reconciles to SAP General Ledger to the cent every day
How Measured	Daily automated comparison of Gold layer financial totals against SAP General Ledger. Variance report generated. If variance is not zero — pipeline stops before Gold layer is updated. Alert sent to Johan with full details for approval or rejection.
Frequency	Daily
Source Systems	SAP ECC General Ledger + Gold layer Finance fact tables

Acceptance Criteria	Johan can check variance between SAP and platform in seconds — not days. When variance is detected Johan receives alert with full details and can approve or reject before Gold layer is updated. Zero variance is the only acceptable Gold layer state.
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FR-FIN-003 — Month-end Close Pack

Requirement ID	FR-FIN-003
What	Automated month-end close pack in same format as current Excel workbooks delivered by 7am on day two after month-end
How Measured	dbt Gold models produce all financial summaries. Excel workbook business rules migrated to dbt transformation models with version control. Same format as current Excel output.
Frequency	Monthly — delivered by 7am on day two after month-end
Source Systems	SAP ECC + dbt Gold models + Excel workbook business logic migrated to dbt transformation models
Acceptance Criteria	Platform output compared to Excel workbooks — numbers must correspond exactly. Report delivered by 7am on day two. Johan and team get three days back. Two consecutive months must reconcile before Excel is decommissioned. Johan signs off each month.

FR-FIN-004 — Creditors Aging Analysis

Requirement ID	FR-FIN-004
What	Three-way match and exception queue visible and managed in platform replacing current Excel-based creditors aging process
How Measured	SAP three-way match — Purchase Order plus GRN plus Invoice. Exceptions flagged with rejection reason and age in queue. Exceptions queue managed in platform not Excel.
Frequency	Weekly
Source Systems	SAP ECC invoice records + Purchase Order records + GRN records
Acceptance Criteria	Platform shows all 847 current exception queue invoices with rejection reason and days in queue. Count matches Excel exceptions queue exactly on first validation run.

FR-FIN-005 — Backdated Posting Alert

Requirement ID	FR-FIN-005
What	Alert sent to Johan within one hour of any backdated SAP posting detected after nightly extraction
How Measured	Real-time Change Data Capture on SAP detects backdated postings made after nightly extraction. Alert contains transaction date, posting date, amount, period affected, and approve or reject workflow. Gold layer not updated until Johan approves.
Frequency	Real-time detection. Alert sent within one hour of detection.
Source Systems	SAP ECC + Change Data Capture + Alert framework
Acceptance Criteria	Johan confirms alert received within one hour of detection. Alert contains sufficient detail to approve or reject without logging into the platform. Gold layer only updated after Johan's approval.

9. Mart 6 — Marketing Attribution

Process Owner: Rethabile Sithole, Chief Commercial Officer. Source: BPA-001 Section 3.3.

BPA-FINDING-002 applies to this entire mart — marketing attribution data does not exist in any RetailSphere source system. The platform creates it for the first time. RetailSphere spends R8.3 million annually on marketing. ADR-006 must be formally agreed and signed by the CCO before any Gold layer marketing build begins. FR-MKT-002, FR-MKT-003, and FR-MKT-004 are decision requirements — they define methodology that Rethabile must sign off on before the platform is built.

FR-MKT-001 — Closed-loop Marketing Attribution

Requirement ID	FR-MKT-001
What	Connect R8.3M annual marketing spend to revenue — show which spend drove which revenue and through which channel
How Measured	Platform links campaign send events to conversion events across all three marketing systems. Single attribution per conversion — no double counting. Closed loop between campaign spend and purchase confirmed.
Frequency	Monthly
Source Systems	Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4 + Oracle MICROS POS + Shopify Plus (conversion)
Acceptance Criteria	Rethabile can see campaign spend connected to revenue regardless of which channel the customer converted on. Double and triple counting eliminated. R8.3M spend attributed to confirmed revenue.
Dependencies	ADR-006 — Marketing Attribution Logic must be signed before build. ADR-002 — Unified Customer Identity required to link touchpoints across channels.
Note	BPA-FINDING-002 — attribution data does not exist in any source system. Platform creates it for the first time.

FR-MKT-002 — Attribution Window Definition

Requirement ID	FR-MKT-002
What	Define the number of days between a campaign touchpoint and a purchase that counts as an attributed conversion
How Measured	Platform calculates time between campaign touchpoint event and subsequent purchase transaction across all channels. Applies agreed window consistently to all attribution calculations.
Frequency	Applied to every campaign send and conversion event
Source Systems	Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4
Acceptance Criteria	Rethabile signs off on the attribution window definition before platform build begins. Window definition documented in ADR-006.
Dependencies	ADR-006 — must be agreed and signed before this is built. Cannot be changed after Rethabile begins making spend decisions based on platform numbers.

FR-MKT-003 — Attribution Model Type

Requirement ID	FR-MKT-003
What	Define which attribution model the platform applies consistently across all three marketing systems
How Measured	Platform applies one agreed attribution model — options include last-click, first-click, linear, time-decay, or data-driven. Applied consistently across Salesforce Marketing Cloud, Meta, and GA4.
Frequency	Applied to every conversion event calculation
Source Systems	Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4
Acceptance Criteria	Rethabile signs off on model type before build begins. Model type documented in ADR-006. Once agreed — cannot be changed after Rethabile begins making R8.3M spend decisions based on platform attribution numbers.
Dependencies	ADR-006 — must be agreed and signed before this is built

FR-MKT-004 — Cross-channel Attribution Weighting

Requirement ID	FR-MKT-004
What	Define how credit is shared when a customer touches multiple channels before converting
How Measured	When a customer engages with multiple channels before converting — platform applies agreed weighting to determine which channel receives credit and how much. Single attribution per conversion — no double counting.
Frequency	Applied to every multi-touch conversion event
Source Systems	Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4 + Shopify Plus
Acceptance Criteria	Rethabile signs off on weighting methodology before build begins. Methodology documented in ADR-006. Double and triple counting eliminated — each conversion attributed once only.
Dependencies	ADR-006 — must be agreed before build. ADR-002 — Unified Customer Identity required to link touchpoints across channels to the same person.

10. Cross-cutting Requirements

Cross-cutting requirements apply across all six marts simultaneously. They are not owned by a single domain — they are owned by the platform architecture.

FR-CCR-001 — Data Lineage Across All Layers

Requirement ID	FR-CCR-001
What	Every number in the Gold layer must be traceable back to its source transaction through Bronze and Silver layers
How Measured	Column-level lineage tracked from source through Bronze, Silver, and Gold. dbt documentation provides lineage diagrams per model.
Frequency	Continuous — maintained as part of every pipeline run
Source Systems	dbt transformation logs + Bronze metadata + Silver metadata + Gold metadata

Acceptance Criteria	Any Gold layer number can be traced to its source transaction within minutes. Auditors and analysts can follow the full journey of any data point.
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FR-CCR-002 — Master Product Identifier

Requirement ID	FR-CCR-002
What	Single trusted product identifier resolving SAP stock code, POS code, and Shopify SKU to one surrogate key
How Measured	Cross-reference table in Silver layer maps all three product identifiers to a single Product dimension surrogate key. All marts join to Product dimension via this key.
Frequency	Updated daily — new products and identifier changes detected and resolved
Source Systems	SAP ECC + Oracle MICROS POS + Shopify Plus
Acceptance Criteria	All six marts reference the same Product dimension surrogate key. SAP-only, POS-only, and Shopify-only product counts reconcile to the unified product count.
Dependencies	ADR-001 — Master Product Identifier Strategy must be resolved before Product dimension can be finalised

FR-CCR-003 — Unified Customer Identifier

Requirement ID	FR-CCR-003
What	Single trusted customer identifier linking Salesforce loyalty ID, Shopify customer ID, and POS receipt to one surrogate key
How Measured	Identity resolution logic in Silver layer. Deterministic matching for loyalty customers using email, name, and address. Segment framework for unmatched population.
Frequency	Updated daily
Source Systems	Salesforce CRM + Shopify Plus + Oracle MICROS POS
Acceptance Criteria	All domains reference the same Customer dimension surrogate key. Cross-channel CLV, retention rate, and at-risk flags all calculated from unified identifier.
Dependencies	ADR-002 — Unified Customer Identity Resolution must be resolved before Customer dimension can be finalised. POPIA legal review required.

FR-CCR-004 — POPIA Compliance

Requirement ID	FR-CCR-004
What	Platform must comply with POPIA across all domains — opt-out propagation, data retention, and right to erasure
How Measured	Opt-out events propagated within four hours across all systems. Data retention rules enforced per domain. Right to erasure workflow built into Customer dimension management.
Frequency	Continuous
Source Systems	All source systems + Customer dimension + Contact History fact table
Acceptance Criteria	Compliance audit can confirm opt-out propagation, data retention adherence, and erasure capabilities within 24 hours of request.
Dependencies	POPIA legal review required for ADR-002 identity resolution approach

FR-CCR-005 — Reconciliation Framework

Requirement ID	FR-CCR-005
What	Every Gold layer financial and operational total must reconcile to source system totals
How Measured	Automated reconciliation checks at every layer boundary — Bronze to Silver, Silver to Gold. Variance reports generated daily. Pipeline stops if reconciliation fails.
Frequency	Daily — at every pipeline run
Source Systems	All source systems + Bronze layer + Silver layer + Gold layer
Acceptance Criteria	No unreconciled variance exists in Gold layer. Any detected variance triggers pipeline stop and alert before Gold layer is updated.

FR-CCR-006 — Error Handling and Alerting

Requirement ID	FR-CCR-006
What	No silent failures — every pipeline error detected, logged, and alerted to the relevant team
How Measured	Detailed alerts with: what failed, where it failed, what data is affected, and recommended resolution steps. Linked to troubleshooting runbook.
Frequency	Real-time detection and alerting
Source Systems	All pipeline components + Azure Data Factory + dbt runs
Acceptance Criteria	Every pipeline failure generates a detailed alert within minutes. No wrong data reaches Gold layer without detection. On-call DBA can diagnose and resolve from alert content alone.

FR-CCR-007 — Data Quality Gates

Requirement ID	FR-CCR-007
What	Pipeline stops at every layer boundary if data quality checks fail — bad data cannot silently reach Gold layer
How Measured	dbt tests run at every model execution — not_null, unique, accepted_values, referential integrity. Pipeline halts and alerts if any test fails.
Frequency	Every pipeline run — Bronze to Silver and Silver to Gold boundaries
Source Systems	All pipeline layers
Acceptance Criteria	No requirement can be met with data that has failed quality checks. Gold layer only updated after all quality gates pass.

FR-CCR-008 — Cross-system Opt-out Propagation

Requirement ID	FR-CCR-008
What	Opt-out events propagated across all four marketing systems within four hours — platform-wide legal obligation
How Measured	Tiered ingestion pipeline detects opt-out event in any system and propagates to all four systems within four-hour window. Confirmation alert sent when propagation is complete.

Frequency	Near real-time — four-hour maximum
Source Systems	Salesforce CRM + Salesforce Marketing Cloud + Meta Business Manager + Google Analytics 4
Acceptance Criteria	Opt-out event confirmed across all four systems within four hours. POPIA compliance audit can confirm propagation timeline for any opt-out event.
Note	POPIA legal obligation. Applies platform-wide — not limited to Customer Analytics mart.

11. Requirements Traceability Matrix

Every functional requirement traces back to a specific stakeholder need in BPA-001. This matrix confirms no requirement was invented by the architect — every one was requested by a domain process owner.

Req ID	Requirement	Stakeholder	BPA-001 Source
FR-PRO-001	Supplier on-time delivery rate	Sipho Dlamini	Section 3.1 — Good in 12 months
FR-PRO-002	Invoice accuracy rate	Sipho Dlamini	Section 3.1 — Questions cannot answer
FR-PRO-003	Price variance vs contracted	Sipho Dlamini	Section 3.1 — Good in 12 months
FR-PRO-004	Days payable outstanding	Sipho Dlamini	Section 3.1 — Good in 12 months
FR-PRO-005	Procurement savings %	Sipho Dlamini	Section 3.1 — Decisions monthly
FR-INV-001	Stock position per SKU per store	Zanele Mokoena	Section 3.2 — Questions cannot answer
FR-INV-002	System vs physical variance	Zanele Mokoena	Section 3.2 — Questions cannot answer
FR-INV-003	Shrinkage by category	Zanele Mokoena	Section 3.2 — Good in 12 months
FR-INV-004	Open replenishment orders	Zanele Mokoena	Section 3.2 — Good in 12 months
FR-INV-005	Average days in pipeline	Zanele Mokoena	Section 3.2 — Good in 12 months
FR-SAL-001	Customer Lifetime Value	Rethabile Sithole	Section 3.3 — Questions cannot answer
FR-SAL-002	True retention and churn rate	Rethabile Sithole	Section 3.3 — Questions cannot answer
FR-SAL-003	Customer segment framework	Rethabile Sithole	Section 3.3 — Good in 12 months
FR-CUS-001	Communications per member	Priya Naidoo	Section 3.4 — Good in 12 months
FR-CUS-002	Opt-out propagation	Priya Naidoo	Section 3.4 — Key business processes
FR-CUS-003	At-risk member flag	Priya Naidoo	Section 3.4 — Good in 12 months
FR-CUS-004	Campaign performance dashboard	Priya Naidoo	Section 3.4 — Decisions weekly

Req ID	Requirement	Stakeholder	BPA-001 Source
FR-CUS-005	Net member movement + churn	Priya Naidoo	Section 3.4 — Decisions monthly
FR-CUS-006	Customer segment history SCD2	Rethabile Sithole	Section 3.3 — Good in 12 months
FR-FIN-001	Audit trail report	Johan van der Merwe	Section 3.5 — Good in 12 months
FR-FIN-002	GL reconciliation	Johan van der Merwe	Section 3.5 — Questions cannot answer
FR-FIN-003	Month-end close pack	Johan van der Merwe	Section 3.5 — Good in 12 months
FR-FIN-004	Creditors aging analysis	Johan van der Merwe	Section 3.5 — Good in 12 months
FR-FIN-005	Backdated posting alert	Johan van der Merwe	Section 3.5 — Key business processes
FR-MKT-001	Closed-loop attribution	Rethabile Sithole	Section 3.3 — Questions cannot answer
FR-MKT-002	Attribution window	Rethabile Sithole	Section 3.3 — ADR-006 dependency
FR-MKT-003	Attribution model type	Rethabile Sithole	Section 3.3 — ADR-006 dependency
FR-MKT-004	Cross-channel weighting	Rethabile Sithole	Section 3.3 — ADR-006 dependency
FR-CCR-001	Data lineage all layers	N. Mahlangu / T. Mbalo	BPA-001 architectural implications
FR-CCR-002	Master product identifier	All domains	ADR-001 — BDR-001 v1.0
FR-CCR-003	Unified customer identifier	All domains	ADR-002 — BDR-001 v1.0
FR-CCR-004	POPIA compliance	Priya Naidoo / Legal	Section 3.4 — POPIA legal obligation
FR-CCR-005	Reconciliation framework	Johan van der Merwe	Section 3.5 — Architectural implications
FR-CCR-006	Error handling and alerting	All domains	BPA-001 cross-domain findings
FR-CCR-007	Data quality gates	All domains	FR-FIN-002 pipeline stop requirement
FR-CCR-008	Cross-system opt-out	Priya Naidoo	FR-CUS-002 — POPIA platform-wide

12. Next Steps

- 1. CDO review and sign-off on FRD-001 v1.0.
- 2. Stakeholder validation — share relevant mart sections with each process owner for confirmation.
- 3. ADR-006 formally written and signed by Rethabile Sithole before any Marketing mart build.
- 4. ADR-002 POPIA legal review initiated — blocks FR-SAL-001, FR-SAL-002, FR-CUS-003, FR-CUS-006, FR-MKT-004.
- 5. Proceed to Step 4 — Non-Functional Requirements Definition.
- 6. Upload FRD-001 to GitHub: docs/03-functional-requirements/FRD-001_v1.0.docx

13. Document Change Log

Version	Date	Author	Description
1.0	9 May 2026	T. Mbalo	Initial release. 36 requirements across 6 marts and 8 cross-cutting. All requirements traced to BPA-001 stakeholder interviews.

— End of Document —

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